

Notes: Brugler & Comerton-Forde (JFE, 2025)

Differential Access to Dark Markets and Execution Outcomes

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1 Big picture

- **Question.** Do access restrictions in dark pools affect execution outcomes, and in particular do dark pools that restrict high-frequency traders or electronic liquidity providers deliver better outcomes?
- **Setting.** The paper studies Australian equity dark trades from January 1, 2017 to September 30, 2019. Australia is useful because broker dark pools can impose access restrictions, while exchange-operated dark pools cannot.
- **Main comparison.** The authors compare trades in broker-operated dark pools with trades in exchange-operated dark pools, especially ASX Centre Point and Cboe hidden orders.
- **Core result.** Broker dark pool trades have less order-flow information leakage, lower adverse-selection risk, and smaller post-trade order imbalances than comparable exchange dark pool trades.
- **Mechanism.** Broker dark pools can segment order flow by restricting or allowing opt-out from HFT/ELP, principal, and aggregator flow. Exchange dark pools cannot impose comparable counterparty restrictions.
- **Identification.** Three broker dark pools close during the sample. After closure, their customers must use exchange dark pools for dark executions. These closures provide plausibly exogenous variation in venue choice.
- **Matching evidence.** Trades in continuing broker dark pools are matched to Centre Point trades from brokers whose pools have closed. Matched estimates confirm the panel results.
- **HFT/ELP evidence.** Within broker dark pools, trades in pools that restrict HFT/ELP access have lower information leakage and lower adverse-selection measures than trades in pools that permit HFT/ELP access.
- **Policy implication.** Access restrictions can benefit investors by allowing them to choose venues with lower information leakage, though this must be balanced against execution probability and fairness concerns.
- **Takeaway.** Dark pools are not homogeneous. Their access rules are an economically important dimension of market design.

2 Data and measurement

2.1 Australian dark pool structure

- The paper distinguishes three dark trading categories:
 - **Broker dark pools:** broker-operated dark pools that can impose access restrictions.
 - **Centre Point:** an ASX exchange dark pool.
 - **Cboe Hidden:** hidden orders on Cboe Australia.
- Exchange dark pools provide unrestricted access to traders and cannot screen counterparties in the same way as broker dark pools.
- Broker dark pools vary in how much HFT/ELP, principal, and aggregator flow they allow.

Interpretation: The institutional contrast is central. Broker pools can segment order flow; exchange dark pools generally cannot.

2.2 Sample and trade identification

- The sample covers all dark trades in a large sample of Australian stocks from January 2017 to September 2019.

- The data identify whether a trade takes place in a broker dark pool, Centre Point, or Cboe Hidden.
- Trades are mostly executed at the midpoint of the National Best Bid and Offer, so trade direction is hard to infer.
- Because midpoint trades do not reveal buyer-initiated versus seller-initiated direction, standard effective spread and realized spread measures are not the main outcomes.

Interpretation: The paper works at the trade and stock-day levels, using post-trade price and order-book outcomes that do not require signing trades.

2.3 Main execution outcomes

The main trade-level outcomes are:

- **Absolute price change:** absolute log change in the NBBO midpoint after the trade.
- **Post-trade bid–ask spread:** NBBO spread after the trade.
- **Post-trade dollar volume:** lit market dollar volume after the trade.
- **Absolute order imbalance:** magnitude of directional trading imbalance after the trade.
- **Quote-to-trade ratio:** number of NBBO quote changes divided by number of trades over the interval.

Interpretation: Absolute price change captures information leakage; post-trade spread captures adverse selection risk; order imbalance and trading activity capture whether the trade induces directional follow-on activity.

2.4 Broker dark pool access classifications

Broker dark pools are classified into:

- **Restricted pools:** pools that do not permit HFT/ELP activity and have the most restrictive access rules.
- **Opt-in-to-restrictions pools:** pools that allow customers to opt out of interacting with certain counterparty types, principal flow, or order-by-order flow.

Interpretation: This classification is used to test whether HFT/ELP access drives the execution-quality differences between broker and exchange dark pools.

2.5 Pool closures

Three broker dark pools close during the sample:

- BAML InstinctX closes in March 2017.
- UBS PIN closes in April 2019.
- Citi Match closes in July 2019.

Interpretation: After closure, customers who still demand dark executions must shift toward exchange dark pools. This creates the natural experiment used in the matching design.

3 Models and empirical specifications

3.1 Trade-level panel specification

The trade-level model is

$$y_{ijt} = \alpha_j + \gamma_t + \beta \text{BDP}_{ijt} + \rho' X_{ijt} + \varepsilon_{ijt}, \quad (1)$$

where y_{ijt} is an execution outcome for trade i in stock j on day t , BDP_{ijt} equals one for broker dark pool trades, α_j are stock fixed effects, and γ_t are date fixed effects.

Interpretation: A negative β for absolute price changes means broker dark pool trades leak less information than exchange dark pool trades. A negative β for post-trade spread means lower adverse-selection risk.

3.2 Stock-day averaged specification

Because trade-level estimation is computationally difficult with hundreds of millions of trades, the paper averages variables to the stock-day level:

$$\bar{y}_{jt} = \alpha_j + \gamma_t + \beta \overline{\text{BDP}}_{jt} + \rho' \bar{X}_{jt} + \bar{\varepsilon}_{jt}. \quad (2)$$

- $\bar{y}_{jt}$ is the stock-day average of the execution outcome.
- $\overline{\text{BDP}}_{jt}$ is the fraction of dark pool trades occurring in broker dark pools.
- Controls are stock-day averages of trade size, price, total volume, lit trades, spread, depth, and related order-book variables.

Interpretation: The averaged regression recovers the same coefficient of interest while making the estimation feasible.

3.3 Matching specification for pool closures

For each closure event, broker dark pool trades from continuing pools are matched to Centre Point trades by closed-pool brokers using:

- stock;
- log trade size;
- log execution price;
- log total dollar volume;
- bid–ask spread;
- depth at NBBO;
- time of day and intraday trend controls.

The estimated treatment effect is the mean difference:

$$ATT = E[y^{BDP} - y^{CP} \mid \text{matched trades}]. \quad (3)$$

Interpretation: The matching design compares broker-pool trades to similar Centre Point trades that plausibly would have been routed to a broker dark pool if the broker pool had remained open.

3.4 HFT/ELP access regression

Within broker dark pool trades, the paper estimates:

$$\bar{y}_{jt} = \alpha_j + \gamma_t + \beta \text{Restricted}_{jt} + \rho' \bar{X}_{jt} + \bar{\epsilon}_{jt}, \quad (4)$$

where Restricted_{jt} measures the fraction of broker dark pool trades executed in pools that restrict HFT/ELP access.

Interpretation: A negative coefficient indicates that HFT/ELP-restricted broker pools have better execution outcomes than other broker pools.

3.5 Execution risk robustness

To test whether execution risk changes venue choice, the paper estimates regressions of total dark market share and broker dark pool share on high volatility indicators:

$$y_{jt} = \alpha_j + \gamma_t + \beta RV_{jt} + \rho' X_{jt} + \epsilon_{jt}, \quad (5)$$

where RV_{jt} equals one when realized volatility is above the upper quartile.

Interpretation: This test asks whether traders switch between broker and exchange dark pools when non-execution risk rises.

4 Identification and empirical design

4.1 Main panel identification

- The panel compares broker dark pool and exchange dark pool trades within stock and date.
- Controls absorb trade size, price, market volume, spread, depth, and other order-book conditions.
- Stock and date fixed effects absorb persistent stock characteristics and market-wide conditions.

Interpretation: The panel estimates are informative but may still suffer from unobserved order-type or customer-composition differences.

4.2 Why pool closures help

- Pool closures are motivated by regulatory and compliance concerns rather than observed execution outcomes.
- After a broker closes its pool, the broker cannot route customers to that dark pool.
- Some trades that would have gone to the broker dark pool now go to exchange dark pools.
- Matching focuses on similar trades and stocks around closure windows.

Interpretation: The closure design gives a more causal comparison than ordinary cross-venue panel regressions.

4.3 Mechanism identification

- Broker dark pools differ in whether they restrict HFT/ELP access.
- If HFT/ELP access causes information leakage, restricted pools should have lower leakage than other broker pools.
- The paper finds this pattern in both absolute price changes and post-trade spreads.

Interpretation: This test links the main broker-versus-exchange differences to trader access restrictions rather than to dark-pool labels alone.

4.4 Limitations

- The paper does not observe the full parent orders behind child trades.
- Trade direction is often unobservable because most dark trades occur at the midpoint.
- Broker customers and order types may differ in ways that are hard to fully control.
- Results are based on the Australian market structure and may not map exactly to all jurisdictions.

Interpretation: The strongest causal evidence comes from the closure-matching design, while the panel and access-restriction regressions provide complementary evidence.

5 Tables and figures

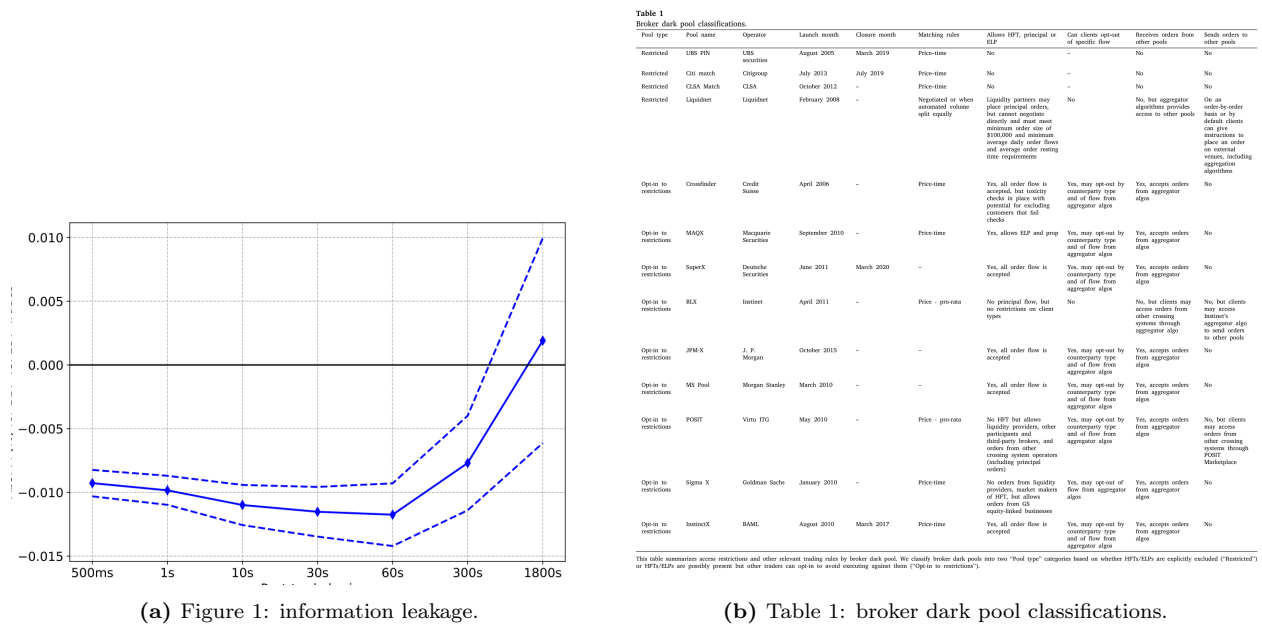
The figures and tables below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and Table 1: information leakage and access restrictions

Read:

- Figure 1 shows estimated differences in absolute post-trade price changes between broker dark pool and exchange dark pool trades.
- The coefficients are negative from 500ms to 300s, meaning broker dark pool trades have lower information leakage.
- The effect is especially visible around 60s, where the broker dark pool coefficient is roughly around -1 basis point.
- Table 1 classifies broker dark pools by access rules, including restricted pools and pools with opt-in restrictions.

Economic message: Broker dark pools are not just another form of dark trading. Their access restrictions can directly reduce the information revealed by a trade.

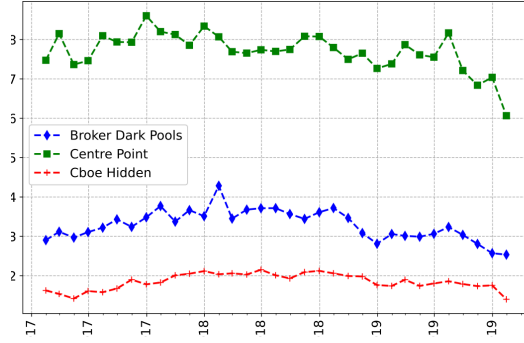


5.2 Figure 2 and Table 2: market shares and summary statistics

Read:

- Figure 2 shows that Centre Point has the largest dark market share, around 6–9% of total trading.
- Broker dark pools account for roughly 2.5–4.5% of trading, while Cboe Hidden is smaller.
- Table 2 reports stock-day summary statistics for dark pool trades.
- Average trade size is about AUD 2,020, and 92% of dark trades occur at the midpoint.

Economic message: Broker dark pools are economically meaningful but smaller than Centre Point. Their smaller share makes the execution-quality differences especially important for routing choices.



(a) Figure 2: dark pool market shares.

Table 2
Stock-day summary statistics.

	Mean	SD	Min	25%	50%	75%	Max
Trade size (AUD '000s)	2.02	13.3	0.00	0.44	1.60	2.10	3.403
Total dollar value (AUD 'm)	13.8	33.0	0.00	0.51	2.99	13.0	1.458
Price (AUD)	8.64	18.8	0.00	1.40	3.38	7.76	241
Daily average dollar depth (AUD '000s)	101	227	1.38	12.5	33.2	95.8	2,126
Daily average bid-ask spread (%)	0.57	0.78	0.02	0.16	0.34	0.61	14.0
Broker dark pool	0.27	0.24	0.00	0.04	0.25	0.43	1.00
Pre-cross bid-ask spread (%)	0.44	0.43	0.00	0.14	0.31	0.51	3.92
Abs. 500ms price change (%)	0.01	0.03	0.00	0.00	0.01	0.01	0.31
Abs. 10s price change (%)	0.03	0.05	0.00	0.01	0.02	0.03	0.48
Abs. 60s price change (%)	0.06	0.08	0.00	0.02	0.04	0.06	0.84
Abs. 1800s price change (%)	0.33	0.31	0.00	0.15	0.24	0.40	3.89
500ms Bid-ask spread (%)	0.45	0.43	0.00	0.15	0.31	0.52	3.92
10s Bid-ask spread (%)	0.45	0.44	0.00	0.15	0.32	0.52	3.92
60s Bid-ask spread (%)	0.47	0.47	0.00	0.16	0.33	0.54	3.92
1800s Bid-ask spread (%)	0.68	0.71	0.00	0.24	0.45	0.80	5.89
60s dollar volume (AUD '000s)	39.2	75.2	0.00	3.72	13.2	42.7	2,018
60s Abs. order imbalance (AUD '000s)	18.0	28.1	0.00	2.68	8.16	21.5	650
60s Quote-to-trade ratio	3.96	1.35	0.06	3.09	3.91	4.72	28.0

This table contains stock-day summary statistics for dark pool trades in stocks in the ASX All Ordinaries Index from the period Jan 1, 2017 to Sep 30, 2019. All stock-days that record at least one trade either on a broker dark pool or an exchange dark pool are included in the sample. All variables are averaged across all dark pool trades by stock-day unless otherwise stated. Trade size is the dollar volume of the trade in thousands of dollars. Total dollar value is the total amount traded summed across all venues for that stock-day in millions of dollars. Price is the average trade price in dollars. Daily Average Bid-ask Spread is the time-weighted average log bid-ask spread at the NBBO for that stock and day. Daily Average Dollar Depth is the time-weighted average depth available at the NBBO for that stock and day in thousands of dollars. Broker Dark Pool is a dummy variable taking the value one if a trade is on a broker dark pool and zero if the trade is in an exchange dark pool, (i.e. it is the average proportion of dark pool trades on broker dark pools). Pre-Cross Bid-ask Spread is the log bid-ask spread at the NBBO at the time of the trade. Absolute τ price change is the absolute log difference between the mid-quote τ seconds after the trade and the

(b) Table 2: stock-day summary statistics.

5.3 Table 3 and Table 4: panel evidence on execution outcomes

Read:

- Table 3 shows that the broker dark pool coefficient is negative for absolute price changes from 500ms through 300s.
- The effect is statistically significant at the short and medium horizons, implying less information leakage.
- Table 4 shows that broker dark pool trades are followed by lower bid-ask spreads at most horizons.
- The broker dark pool spread coefficients are negative and significant through 60s.

Economic message: Broker dark pool trades reduce both information leakage and adverse-selection risk relative to exchange dark pool trades.

Table 3

Information leakage panel regressions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	500ms	1s	10s	30s	60s	300s	1800s
Ln(Dollar trade size)	-0.0009 (-5.19)	-0.0008 (-4.21)	-0.0005 (-2.10)	-0.0006 (-2.06)	-0.0007 (-1.86)	-0.0020 (-3.74)	-0.0046 (-4.49)
Ln($N_{darkpool}$)	-0.0083 (-33.4)	-0.0094 (-33.1)	-0.0131 (-35.6)	-0.0167 (-36.9)	-0.0203 (-36.4)	-0.0315 (-36.3)	-0.0478 (-37.8)
Ln(N_{lit})	0.0056 (17.9)	0.0064 (17.6)	0.0111 (21.9)	0.0158 (25.7)	0.0215 (27.4)	0.0472 (33.2)	0.1026 (34.8)
Ln(Price)	-0.0001 (-0.18)	0.0002 (0.23)	-0.0009 (-0.69)	-0.0043 (-2.50)	-0.0092 (-4.05)	-0.0406 (-9.32)	-0.1422 (-13.9)
Ln(Dollar volume)	0.0032 (14.5)	0.0035 (13.6)	0.0062 (16.1)	0.0098 (19.7)	0.0138 (20.1)	0.0317 (22.6)	0.0720 (23.2)
Pre-cross bid-ask spread	0.0069 (5.95)	0.0098 (7.33)	0.0192 (9.81)	0.0262 (10.3)	0.0367 (10.8)	0.0798 (12.8)	0.1808 (15.6)
Ln(Depth)	-0.0047 (-19.2)	-0.0056 (-20.1)	-0.0102 (-25.0)	-0.0144 (-27.7)	-0.0189 (-29.3)	-0.0360 (-29.3)	-0.0699 (-27.5)
Broker dark pool	-0.0093 (-17.5)	-0.0098 (-16.9)	-0.0110 (-13.7)	-0.0115 (-11.6)	-0.0117 (-9.39)	-0.0077 (-4.08)	0.0019 (0.47)
Fixed effects	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$
R^2	0.08	0.08	0.09	0.10	0.11	0.13	0.13
N_{obs}	242,840	242,840	242,840	242,840	242,840	242,840	242,840
N_{stocks}	626	626	626	626	626	626	626

This table contains estimates from regressions of the stock-day average post-trade absolute price change at various horizons after a dark pool trade onto stock-day level controls, fixed effects and the fraction of all dark pool trades that occur on a broker dark pool. The regression model is $\bar{y}_{jt} = \alpha_j + \gamma_t + \beta BDP_{jt} + \rho' \bar{X}_{jt} + \varepsilon_{jt}$ where α_j is a stock fixed effect, γ_t is a date fixed effect, $\bar{y}_{jt}$ is the stock-day average of the absolute impact after a trade on either an exchange dark pool or a broker dark pool for stock j and day t , $\bar{X}_{jt}$ is the stock-day average of a vector of controls including: log of dollar trade size, log of trade price, bid-ask spread and log depth available at the NBBO at the time of the trade, the log of total dollar volume traded across all trades and venues, and the log of total number of dark pool and lit trades; BDP_{jt} is the stock-day average of a dummy variable that takes the value 1 if the trade occurs on a broker dark pool and 0 otherwise (i.e. the fraction of broker dark pool trades out of all dark pool trades) and ε_{jt} is an error term. We estimate the model for horizons of 500ms, 1s, 10s, 30s, 60s, 300s and 1800s using all stock-days from Jan 1, 2017 to Sept 30, 2019. Reported R^2 values relate to the within variation in the dependent variables. Standard errors are clustered at the stock level and t -statistics are in parentheses.

Table 3: information leakage panel regressions

Adverse selection panel regressions.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	500ms	1s	10s	30s	60s	300s	1800s
Ln(Dollar trade size)	-0.0010 (-3.38)	-0.0012 (-4.07)	-0.0017 (-5.67)	-0.0023 (-6.89)	-0.0025 (-5.89)	-0.0042 (-4.78)	-0.0123 (-5.82)
Ln($N_{darkpool}$)	-0.0004 (-1.08)	-0.0010 (-2.39)	-0.0024 (-4.97)	-0.0041 (-7.45)	-0.0058 (-8.52)	-0.0098 (-8.47)	-0.0065 (-2.72)
Ln(N_{lit})	-0.0043 (-6.49)	-0.0045 (-6.60)	-0.0052 (-6.67)	-0.0058 (-6.60)	-0.0068 (-6.10)	-0.0153 (-6.38)	-0.0304 (-5.87)
Ln(Price)	-0.0198 (-8.68)	-0.0210 (-8.83)	-0.0233 (-8.25)	-0.0257 (-7.55)	-0.0293 (-5.84)	-0.0622 (-5.09)	-0.1546 (-6.00)
Ln(dollar volume)	0.0018 (3.27)	0.0024 (4.49)	0.0034 (6.02)	0.0041 (6.51)	0.0047 (5.98)	0.0066 (4.07)	0.0125 (3.72)
Pre-cross bid-ask Spread	0.9209 (178)	0.9169 (164)	0.9073 (141)	0.9065 (126)	0.9251 (108)	0.9475 (55.5)	0.8661 (31.7)
Ln(Depth)	-0.0046 (-7.55)	-0.0051 (-8.11)	-0.0059 (-8.56)	-0.0067 (-7.12)	-0.0058 (-3.84)	-0.0018 (-0.51)	0.0027 (0.42)
Broker dark pool	-0.0024 (-2.42)	-0.0033 (-2.82)	-0.0049 (-3.58)	-0.0066 (-4.17)	-0.0053 (-2.96)	-0.0052 (-1.41)	0.0128 (1.72)
Fixed effects	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$	$N \& T$
R^2	0.88	0.87	0.85	0.81	0.75	0.55	0.21
N_{obs}	242,840	242,840	242,840	242,840	242,840	242,840	242,840
N_{stocks}	626	626	626	626	626	626	626

This table contains estimates from regressions of the stock-day average bid-ask spread at various horizons after a dark pool trade onto stock-day level controls, fixed effects and the fraction of all dark pool trades that occur on a broker dark pool. The regression model is $\bar{y}_{jt} = \alpha_j + \gamma_t + \beta BDP_{jt} + \rho' \bar{X}_{jt} + \varepsilon_{jt}$ where α_j is a stock fixed effect, γ_t is a date fixed effect, $\bar{y}_{jt}$ is the stock-day average of the bid-ask spread after a trade on either an exchange dark pool or a broker dark pool for stock j and day t , $\bar{X}_{jt}$ is the stock-day average of a vector of controls including: log of dollar trade size, log of trade price, bid-ask spread and log depth available at the NBBO at the time of the trade, the log of total dollar volume traded across all trades and venues, and the log of total number of dark pool and lit trades; BDP_{jt} is the stock-day average of a dummy variable that takes the value 1 if the trade occurs on a broker dark pool and 0 otherwise (i.e. the fraction of broker dark pool trades out of all dark pool trades) and ε_{jt} is an error term. We estimate the model for horizons of 500ms, 1s, 10s, 30s, 60s, 300s and 1800s using all stock-days from Jan 1, 2017 to Sept 30, 2019. Reported R^2 values relate to the within variation in the dependent variables. Standard errors are clustered at the stock level and t -statistics are in parentheses.

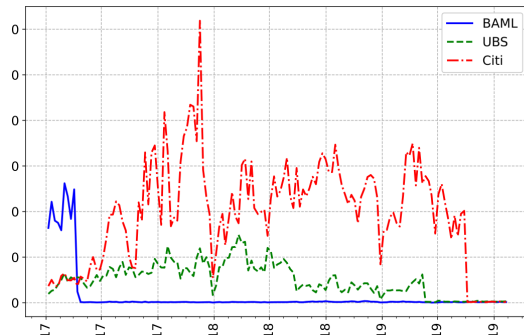
Table 4: adverse selection panel regressions

5.4 Figure 3, Table 5, and Table 6: broker pool closures and matching

Read:

- Figure 3 shows trade counts for BAML, UBS, and Citi dark pools and the dates when those pools close.
- Table 5 reports trade-level summary statistics in closure windows.
- Table 6 gives matching estimates after the three broker pool closures.
- For BAML, UBS, and Citi, matched broker dark pool trades have lower 60s absolute price changes and lower 60s bid-ask spreads than comparable Centre Point trades.

Economic message: The closure-matching evidence supports a causal interpretation: when trades are pushed from broker dark pools toward exchange dark pools, execution outcomes worsen.



(a) Figure 3: closing broker pool trades.

Time-series of the number of broker dark pool trades for each of the three brokers whose pools are closed during our sample period: Bank of America Merrill Lynch (BAML), UBS and Citigroup (Citi). The respective closure dates are March 6, 2017 (BAML), April 1, 2019 (UBS) and July 1, 2019 (Citi).

Table 5
Trade-level summary statistics around pool closure.

	Mean	SD	Min	25%	50%	75%	Max	Mean BDP	Mean CP
Trade size (AUD '000s)	1.79	29.8	0.00	0.08	0.31	1.14	22,025	1.62	2.70
Price (AUD)	20.2	33.0	0.06	4.03	8.73	21.0	232	20.6	18.8
Total dollar value (AUD 'm)	40.6	54.0	0.00	9.07	21.3	46.2	441	40.6	41.0
Pre-cross bid-ask spread (%)	0.18	0.18	0.00	0.06	0.12	0.25	1.87	0.17	0.19
Abs. 500ms price change (%)	0.00	0.02	0.00	0.00	0.00	0.00	0.25	0.00	0.01
Abs. 10s price change (%)	0.01	0.05	0.00	0.00	0.00	0.00	0.42	0.01	0.02
Abs. 60s price change (%)	0.04	0.08	0.00	0.00	0.00	0.04	0.66	0.04	0.04
Abs. 1800s price change (%)	0.24	0.30	0.00	0.03	0.15	0.32	2.79	0.24	0.23
500ms Bid-ask spread (%)	0.18	0.18	0.00	0.06	0.12	0.25	1.87	0.17	0.20
10s Bid-ask spread (%)	0.18	0.18	0.00	0.06	0.12	0.25	1.87	0.17	0.20
60s Bid-ask spread (%)	0.18	0.19	0.00	0.06	0.12	0.26	1.90	0.18	0.21
1800s Bid-ask spread (%)	0.30	0.48	0.00	0.07	0.16	0.32	5.33	0.30	0.30
60s Dollar volume (AUD '000s)	96.5	198	0.00	5.38	24.4	91.9	1,867	94.7	105
60s Abs. order imbalance (AUD '000s)	39.4	84.7	0.00	2.28	10.2	36.6	996	38.7	43.0
60s Quote-to-trade Ratio	3.92	3.15	0.00	2.25	3.04	4.36	28.0	3.95	3.77

This table contains summary statistics for dark pool trades in stocks in the ASX All Ordinaries Index over three one-month periods corresponding to the month after the closure of three broker dark pools, operated by Merrill Lynch (March 6, 2017), UBS (April 1, 2019) and Citigroup (July 1, 2019) respectively. Dark

(b) Table 5: closure-window summary statistics.

Table 6
Matching regression.

	(1) BAML	(2) UBS	(3) Citi
Abs. Price change (60s)	-0.0083 (-8.54)	-0.0037 (-7.84)	-0.0025 (-3.92)
Bid-ask spread (60s)	-0.0264 (-13.6)	-0.0016 (-3.94)	-0.0023 (-4.53)

This table contains average treatment effects obtained from comparing execution outcomes for broker dark pool trades with matched dark pool trades from Centre Point from brokers whose dark pools have recently closed. For each stock and each of the one-month periods after the three pool closures described in Table 5, we match each broker dark pool trade to a Centre Point trade from the closing broker, using propensity score matching on log trade size, log trade price, log total dollar volume traded, bid-ask spread and log depth at NBBO at the time of trade, date, time of day, and quadratic terms for date and time of day, keeping only trades that can be matched using a caliper of 0.25 standard deviations of the estimated propensity scores. We estimate the average treatment effect as the difference in means of the matched broker dark pool trades and Centre Point trades. Treatment effects by event are contained in Columns 1–3 with *t*-statistics from a test that the average effect across stocks is equal to zero in parenthesis below the estimated effect. Execution outcomes are measured using the same variables as defined in Table

5.5 Table 7 and Table 8: HFT/ELP access and post-trade activity

Read:

- Table 7 splits broker dark pool trades by whether pools restrict HFT/ELP access.
- The HFT-restricted pool coefficient is about -0.0080 for absolute price changes and about -0.0037 for spreads.
- Table 8 shows that broker dark pool trades are followed by lower lit market dollar volume and smaller order imbalances than exchange dark pool trades.
- These results are consistent with reduced order anticipation and reduced leakage to fast traders.

Economic message: HFT/ELP access restrictions are a key mechanism. Segmentation reduces the probability that institutional orders interact with counterparties that exploit order-flow information.

Table 7
Stock-day regression for broker dark pool trades split by HFT/ELP access.

	(1) Abs. ΔP	(2) Spread
Ln(Dollar trade size)	0.0029 (10.8)	-0.0003 (-1.37)
Ln($N_{Darkpool}$)	-0.0081 (-23.2)	-0.0033 (-10.1)
Ln(N_{Lit})	0.0171 (21.6)	-0.0060 (-6.78)
Ln(Price)	-0.0115 (-4.80)	-0.0380 (-10.5)
Ln(Dollar volume)	0.0116 (18.8)	0.0033 (5.87)
Pre-cross bid-ask spread	0.0465 (7.86)	0.8744 (115)
Ln(Depth)	-0.0192 (-30.6)	-0.0074 (-9.48)
HFT-restricted pool	-0.0080 (-3.94)	-0.0037 (-1.69)
Fixed effects	N&T	N&T
R^2	0.10	0.77
N	107,068	107,068

(a) Table 7: HFT/ELP access restrictions.

Table 8
Stock-day regression for trading and quoting activity.

	(1) Dollar volume	(2) Abs. order imbalance	(3) Quote to trade
Ln(Dollar trade size)	3.2084 (15.9)	1.8281 (18.9)	0.0253 (5.11)
Ln($N_{Darkpool}$)	-1.8471 (-8.79)	-1.3563 (-16.2)	-0.4029 (-39.8)
Ln(N_{Lit})	7.4024 (10.3)	2.1553 (9.91)	0.3492 (21.3)
Ln(Price)	1.5064 (0.85)	0.0490 (0.08)	0.3613 (6.91)
Ln(Dollar volume)	10.597 (15.3)	5.2233 (18.6)	-0.0649 (-7.06)
Pre-cross bid-ask spread	10.687 (6.47)	3.0117 (5.19)	0.1305 (2.17)
Ln(Depth)	1.7662 (4.54)	1.3657 (9.00)	0.0318 (2.19)
Broker dark pool	-4.9707 (-5.76)	-1.9637 (-7.64)	0.5856 (26.5)
Fixed effects	N&T	N&T	N&T
R^2	0.13	0.17	0.09
N_{obs}	237,180	237,180	237,180
N_{stocks}	624	624	624

This table contains estimates from regressions of the stock-day average of trading and quoting activity after a dark pool trade onto stock-day level controls, fixed effects and the fraction of all dark pool trades that occur on a broker dark pool. The general form of the regression model is $y_{jt} = \alpha_j + \gamma_t + \beta_1 BDP_{jt} + \beta_2 X_{jt} + \epsilon_{jt}$ where α_j is a stock fixed effect, γ_t is a date fixed effect, y_{jt} is the stock-day average of the total dollar volume traded in thousands of dollars (Column 1), the absolute order imbalance (Column 2) in thousands of dollars, or the quote-to-trade ratio (Column 3) for stock j and day t . X_{jt} is the stock-day average of a vector of controls including: log of dollar trade size, log of trade price, bid-ask spread and log depth available at the NBBO at the time of the trade, the log of

(b) Table 8: post-trade trading and quoting activity.

5.6 Table 9: execution risk and routing

Read:

- Table 9 tests whether high realized volatility changes dark pool market share and broker dark pool share.
- High volatility reduces total dark pool share.
- There is limited evidence that high volatility systematically shifts dark trading toward broker dark pools.
- This weakens the idea that the main results are driven by endogenous venue choice during risky periods.

Economic message: Access restrictions improve outcomes, but traders do not appear to strongly substitute toward broker dark pools when execution risk rises.

Table 9
Dark pool trading volume and execution risk.

	Dark fraction of total				BDP fraction of all dark			
	(1)		(2)		(3)		(4)	
Ln(Dollar trade size)	–	–	0.0622	(41.4)	–	–	–0.0208	(–15.0)
Ln($N_{darkpool}$)	–	–	0.0747	(57.6)	–	–	0.0092	(6.08)
Ln(N_{lit})	–	–	–0.0873	(–50.4)	–	–	0.0104	(4.68)
Ln(Price)	–	–	–0.0025	(–1.04)	–	–	0.0513	(7.82)
Ln(Dollar volume)	–	–	0.0069	(5.87)	–	–	0.0033	(2.54)
Pre-cross bid–ask spread	–	–	0.0016	(0.53)	–	–	0.0221	(2.78)
Ln(Depth)	–	–	–0.0244	(–20.4)	–	–	–0.0041	(–2.64)
Upper quartile RV	–0.0391	(–23.7)	–0.0289	(–24.0)	0.0043	(2.00)	0.0006	(0.32)
Fixed effects	$N \& T$		$N \& T$		$N \& T$		$N \& T$	
R^2	0.01		0.48		0.00		0.02	
N_{obs}	242,854		242,854		242,854		242,854	
N_{stocks}	626		626		626		626	

This table contains estimates from regressions of the fraction of total trading that takes place in all dark pools and the fraction of dark pool trading that takes place on broker dark pools onto stock-day level controls, fixed effects and a dummy variable indicating whether stock-day realized volatility is above the upper quartile computed across all stock-days in the month. The general form of the regression model is $y_{jt} = \alpha_j + \gamma_t + \beta RV_{jt} + \rho' X_{jt} + \varepsilon_{jt}$ where α_j is a stock fixed effect, γ_t is a date fixed effect, y_{jt} is the ratio of dark pool volume traded to total volume traded during the continuous trading period, excluding auctions and block trading, (Columns 1 and 2) or the ratio of broker dark pool trading to total dark pool trading (Columns 3 and 4) for stock j and day t , X_{jt} is the stock-day

6 Short summary

- The paper studies how dark pool access restrictions affect execution outcomes.
- The setting is Australia, where broker dark pools can restrict access but exchange dark pools cannot.
- Broker dark pool trades have lower information leakage than exchange dark pool trades.
- Broker dark pool trades also have lower adverse-selection risk and lower post-trade order imbalances.
- A panel design with stock and date fixed effects supports these differences.
- A closure-matching design using three broker dark pool closures supports a causal interpretation.
- Within broker dark pools, restricted pools that limit HFT/ELP access have better execution outcomes.
- The results imply that dark pool segmentation can benefit investors by reducing information leakage.
- The policy trade-off is between fairness and execution probability on one side and information leakage and execution outcomes on the other.

7 Conclusion

7.1 Core conclusion

Brugler and Comerton-Forde show that differential access to dark markets matters. Broker dark pools with access restrictions produce better execution outcomes than unrestricted exchange dark pools. These differences are not merely descriptive; they are supported by evidence from broker dark pool closures.

7.2 Economic intuition

Institutional trades reveal information about parent orders. If a dark pool allows HFTs, ELPs, principal desks, or aggregators to interact freely with this order flow, counterparties may infer trading intentions and trade ahead or around the institutional order. Access restrictions reduce this information leakage by controlling who can interact with the flow.

7.3 Takeaway

The paper reframes dark pool design as a segmentation problem. Access restrictions can be valuable because they let investors choose venues with lower leakage and adverse selection. Regulation that bans such restrictions removes this choice, while regulation that allows restrictions with disclosure may support better execution decisions.